

Projection Certificates and Global-Direction Guarantees for Multivariate Max-Affine Route Risk

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1 Purpose and current status

This note isolates the mathematical mechanism behind a certified acceleration scheme for stochastic pickup-and-delivery routing under multivariate affine demand uncertainty. The current implementation has three properties:

1. a projected one-dimensional CVaR is evaluated by the existing exact $O(p \log N)$ routine;
2. an exact residual radius produces a deterministic two-sided CVaR bracket;
3. every ambiguous route is sent to the full multivariate evaluator.

Therefore, the screening layer changes computational cost but not route feasibility decisions. In the current 40-instance benchmark, the certified solver reproduced the same final plan and objective as the full evaluator, with a median solver speed-up of approximately $2.92\times$.

The remaining mathematical issue is the choice of a single projection direction that is reused across many route checks. The results below provide a deterministic, population-level guarantee for this global direction.

2 Multivariate max-affine route loss and exact certificate

For a route r , let

$$P_r(X) = \max_{0 \leq t < K_r} (a_{rt} + b_{rt}^\top X), \quad X \in \mathbb{R}^d, \quad (1)$$

where K_r is the number of cut slopes. Throughout the empirical analysis, the center is the sample mean

$$\mu := \frac{1}{N} \sum_{s=1}^N X_s. \quad (2)$$

Fix a unit direction $u \in \mathbb{R}^d$. Write each scenario as

$$X_s = \mu + uz_s + e_s, \quad u^\top e_s = 0, \quad (3)$$

with $z_s = u^\top (X_s - \mu)$. The projected route loss is

$$\tilde{P}_{r,u}(z) = \max_{0 \leq t < K_r} (a_{rt} + b_{rt}^\top \mu + (b_{rt}^\top u)z). \quad (4)$$

Throughout, empirical CVaR means the Rockafellar–Uryasev functional

$$\widehat{\text{CVaR}}_\alpha(Y) := \min_{\eta \in \mathbb{R}} \left\{ \eta + \frac{1}{(1-\alpha)N} \sum_{s=1}^N (Y_s - \eta)_+ \right\}, \quad 0 \leq \alpha < 1. \quad (5)$$

Equivalently, it is the upper-tail average with the appropriate fractional weight on the boundary order statistic when $(1 - \alpha)N$ is nonintegral. This empirical functional is coherent, so monotonicity, positive homogeneity, and subadditivity apply below.

Define the exact route residual radius

$$\gamma_r(u) := \max_{0 \leq t < K_r} \|(I - uu^\top)b_{rt}\|. \quad (6)$$

It is computed exactly in $O(K_r d)$ time.

Lemma 2.1 (Pathwise projection error). *For every scenario s ,*

$$\left| P_r(X_s) - \tilde{P}_{r,u}(z_s) \right| \leq \gamma_r(u) \|e_s\|. \quad (7)$$

Proof. For every cut t ,

$$a_{rt} + b_{rt}^\top X_s = a_{rt} + b_{rt}^\top \mu + (b_{rt}^\top u)z_s + ((I - uu^\top)b_{rt})^\top e_s.$$

The last term has absolute value at most $\gamma_r(u)\|e_s\|$. More explicitly, if $|f_t - g_t| \leq \varepsilon$ for every t , then $\max_t f_t \leq \max_t g_t + \varepsilon$ and $\max_t g_t \leq \max_t f_t + \varepsilon$; hence $|\max_t f_t - \max_t g_t| \leq \varepsilon$. Applying this with $\varepsilon = \gamma_r(u)\|e_s\|$ proves the claim. $\square$

Let $C_e(u) := \widehat{\text{CVaR}}_\alpha(\|e\|)$ under the empirical scenario distribution.

Proposition 2.2 (Exact empirical CVaR bracket). *For every route r and unit direction u ,*

$$\widehat{\text{CVaR}}_\alpha(\tilde{P}_{r,u}) - \gamma_r(u)C_e(u) \leq \widehat{\text{CVaR}}_\alpha(P_r) \leq \widehat{\text{CVaR}}_\alpha(\tilde{P}_{r,u}) + \gamma_r(u)C_e(u). \quad (8)$$

Proof. Equation (7) gives

$$\tilde{P}_{r,u} - \gamma_r(u)\|e\| \leq P_r \leq \tilde{P}_{r,u} + \gamma_r(u)\|e\|.$$

Apply monotonicity, positive homogeneity, and subadditivity of empirical CVaR. For the lower bound, use $\text{CVaR}(A) \leq \text{CVaR}(A - B) + \text{CVaR}(B)$ with $A = \tilde{P}_{r,u}$ and $B = \gamma_r(u)\|e\|$. $\square$

For a capacity threshold Q , the route is certified feasible if the upper endpoint in (8) is at most Q , certified infeasible if the lower endpoint exceeds Q , and otherwise evaluated by the exact full multivariate routine. This fallback makes the screening procedure decision-exact.¹

Proposition 2.3 (Direction-free certification margin). *Fix a unit direction u and a route r , and write*

$$h_r(u) := \gamma_r(u)C_e(u) \quad (9)$$

for the half-width in (8). If

$$\left| Q - \widehat{\text{CVaR}}_\alpha(P_r) \right| > 2h_r(u), \quad (10)$$

then, in exact arithmetic, route r is certified, feasible or infeasible according to the sign of $Q - \widehat{\text{CVaR}}_\alpha(P_r)$; the implementation-level padding is described in the footnote above.

¹The implementation adds a conservative floating-point padding band around each endpoint. The padding can only move a route from certified to ambiguous, never reverse a certified decision; the ambiguous case is then resolved by the full evaluator.

Proof. Let $c_r = \widehat{\text{CVaR}}_\alpha(P_r)$ and $\tilde{c}_{r,u} = \widehat{\text{CVaR}}_\alpha(\tilde{P}_{r,u})$. The bracket implies $|c_r - \tilde{c}_{r,u}| \leq h_r(u)$. Hence its upper and lower endpoints satisfy

$$U = \tilde{c}_{r,u} + h_r(u) \leq c_r + 2h_r(u), \quad L = \tilde{c}_{r,u} - h_r(u) \geq c_r - 2h_r(u).$$

If $Q - c_r > 2h_r(u)$, then $U < Q$ and the route is certified feasible. If $c_r - Q > 2h_r(u)$, then $L > Q$ and the route is certified infeasible. $\square$

The left-hand side of (10) depends on the exact route risk and the capacity threshold, but not on the projection direction. Thus reducing $h_r(u)$ makes this sufficient condition easier to satisfy. The condition is not necessary: the bracket may certify additional routes. At population level, the objective introduced below is exactly a weighted squared aggregate of these route half-widths, but it is not claimed to maximize the discontinuous finite-population coverage count.

3 Choosing one global direction

Let $\mathcal{R}$ be a finite population of routes, with positive route weights w_r . For a unit direction u , define

$$S_r(u) := \sum_{t=0}^{K_r-1} \|(I - uu^\top)b_{rt}\|^2, \quad (11)$$

$$\gamma_r(u) := \max_{0 \leq t < K_r} \|(I - uu^\top)b_{rt}\|, \quad (12)$$

and the residual coherence

$$\mu_r(u) := \frac{K_r \gamma_r(u)^2}{S_r(u)} \quad \text{when } S_r(u) > 0. \quad (13)$$

When $S_r(u) = 0$, we set $\mu_r(u) = 1$; this is only a notational convention, since then also $\gamma_r(u) = 0$ and

$$K_r \gamma_r(u)^2 = \mu_r(u) S_r(u) = 0. \quad (14)$$

Define the weighted slope moment and the two population objectives

$$M := \sum_{r \in \mathcal{R}} w_r \sum_{t=0}^{K_r-1} b_{rt} b_{rt}^\top, \quad (15)$$

$$J_S(u) := \sum_{r \in \mathcal{R}} w_r S_r(u), \quad (16)$$

$$J_\gamma(u) := \sum_{r \in \mathcal{R}} w_r K_r \gamma_r(u)^2. \quad (17)$$

The quantity $G_\gamma(u) := \sqrt{J_\gamma(u)}$ is the $w_r K_r$ -weighted RMS exact route radius.

Lemma 3.1 (Residual coherence). *For every route and every unit direction,*

$$1 \leq \mu_r(u) \leq K_r. \quad (18)$$

Both inequalities are tight.

Proof. Let $v_t = \|(I - uu^\top)b_{rt}\|$. Then $\gamma_r^2 = \max_t v_t^2$ and $S_r = \sum_t v_t^2$, so

$$\gamma_r^2 \leq S_r \leq K_r \gamma_r^2.$$

If $S_r > 0$, division yields (18); the zero case follows from the convention. Equality $\mu_r = 1$ occurs when all projected norms are equal, while $\mu_r = K_r$ occurs when exactly one projected norm is nonzero. $\square$

Proposition 3.2 (Second-moment optimizer). *For every unit u ,*

$$J_S(u) = \text{tr}(M) - u^\top M u. \quad (19)$$

Consequently,

$$\arg \min_{\|u\|=1} J_S(u) = \mathcal{E}_1 \cap S^{d-1}, \quad (20)$$

where $\mathcal{E}_1$ is the leading eigenspace of M .

Proof. For every vector b , $\|(I - uu^\top)b\|^2 = \|b\|^2 - (u^\top b)^2$. Summing gives (19). Minimizing J_S is therefore equivalent to maximizing the Rayleigh quotient $u^\top M u$, whose maximizers are exactly the unit vectors in the leading eigenspace. $\square$

Let $u_S \in \arg \min_{\|u\|=1} J_S(u)$ and

$$J_\gamma^\star := \min_{\|u\|=1} J_\gamma(u). \quad (21)$$

The minimum is attained because each γ_r is the maximum of finitely many continuous functions and the unit sphere is compact. The comparator $J_\gamma^\star$ is the best *single global direction for the slope-radius objective*, not a collection of route-specific directions.

Let u_T be a fixed direction learned from a training route population but evaluated on the search population $\mathcal{R}$. The proof below uses no property of the training procedure: u_T may be any fixed unit vector. Assume $J_S(u_S) > 0$, and define

$$\bar{\mu}_T := \max_{r \in \mathcal{R}} \mu_r(u_T), \quad R_M := \frac{J_S(u_T)}{J_S(u_S)}. \quad (22)$$

By the second-moment characterization above, u_S minimizes J_S , and therefore $R_M \geq 1$.

Theorem 3.3 (Global-direction coherence guarantee). *Under the preceding assumptions,*

$$\boxed{J_\gamma(u_T) \leq \bar{\mu}_T R_M J_\gamma^\star.} \quad (23)$$

Equivalently,

$$\boxed{G_\gamma(u_T) \leq \sqrt{\bar{\mu}_T R_M} G_\gamma^\star.} \quad (24)$$

Proof. Using (14),

$$J_\gamma(u_T) = \sum_r w_r \mu_r(u_T) S_r(u_T) \leq \bar{\mu}_T J_S(u_T) = \bar{\mu}_T R_M J_S(u_S).$$

For any unit v , the inequality $S_r(v) \leq K_r \gamma_r(v)^2$ gives $J_S(v) \leq J_\gamma(v)$. Therefore, if $v^\star$ minimizes J_γ ,

$$J_S(u_S) = \min_v J_S(v) \leq J_S(v^\star) \leq J_\gamma(v^\star) = J_\gamma^\star.$$

Combining the two displays proves (23); taking square roots gives (24). $\square$

The theorem separates two losses:

$$\underbrace{R_M}_{\text{training-to-search transfer for } J_S} \times \underbrace{\bar{\mu}_T}_{\text{conversion from } J_S \text{ to } J_\gamma}. \quad (25)$$

Neither factor can be removed.

Proposition 3.4 (Joint tightness and necessity). *Theorem 3.3 is jointly tight with $\bar{\mu}_T > 1$ and $R_M > 1$. The same example shows that neither factor can be dropped.*

Proof. Take one route in $\mathbb{R}^3$ with $K = 2$, $w = 1$, $b_0 = e_1$, and $b_1 = e_2$. Then $M = \text{diag}(1, 1, 0)$ and $J_S(u_S) = 1$. Choose

$$u_T = \cos \theta e_1 + \sin \theta e_3, \quad 0 < \theta < \frac{\pi}{2}.$$

The projected squared norms are $\sin^2 \theta$ and 1, so

$$J_\gamma(u_T) = 2, \quad \bar{\mu}_T = \frac{2}{1 + \sin^2 \theta}, \quad R_M = 1 + \sin^2 \theta.$$

Moreover, $J_\gamma^\star = 1$, attained when $u_1^2 = u_2^2 = 1/2$. Thus

$$\bar{\mu}_T R_M J_\gamma^\star = 2 = J_\gamma(u_T).$$

Since $\bar{\mu}_T < 2$ and $R_M < 2$, dropping either factor makes the right-hand side strictly smaller than $J_\gamma(u_T)$. $\square$

Remark 3.5 (Why the coherence factor cannot be dropped). The direction u_S minimizes J_S , not J_γ . Consequently, $R_M = 1$ does not imply $J_\gamma(u_T) = J_\gamma^\star$, even when the leading eigenvalue is simple. For a single route in $\mathbb{R}^2$ with $K = 2$, $b_0 = 2e_1$, and $b_1 = e_2$, the moment $M = \text{diag}(4, 1)$ has the unique second-moment optimizer $u_S = e_1$. Taking $u_T = u_S$ gives $R_M = 1$ and $J_\gamma(u_T) = 2$, whereas $J_\gamma^\star = 8/5$ is attained when $\sin^2 \phi = 1/5$. Thus the ratio is $5/4$, and the coherence factor is still needed to cover the surrogate-to-radius gap.

Multiplicity creates a second, distinct effect. The surrogate J_S is constant on the leading eigenspace, whereas J_γ need not be. For $b_0 = e_1$ and $b_1 = e_2$, $M = \text{diag}(1, 1, 0)$ and $J_S \equiv 1$ on $\text{span}\{e_1, e_2\} \cap S^2$, while J_γ ranges from 1 to 2 there. In the current diagnostic, all 40 realized search moments were numerically simple, with relative eigengap $1 - \lambda_2(M)/\lambda_1(M) \geq 0.876$, so this multiplicity effect was not active; the first effect remains present regardless of multiplicity.

4 From slope radius to certificate half-width

Theorem 3.3 controls the slope-radius objective J_γ . The actual half-width of the route certificate in (8) is $\gamma_r(u)C_e(u)$, and the residual factor $C_e(u)$ also depends on the direction. Because $C_e(u)$ is common to all routes, the corresponding population objective is

$$\Psi(u) := C_e(u)^2 J_\gamma(u), \quad \sqrt{\Psi(u)} = C_e(u) G_\gamma(u). \quad (26)$$

Since $h_r(u) = \gamma_r(u)C_e(u)$, this objective is exactly the weighted squared aggregate of the route half-widths:

$$\boxed{\Psi(u) = \sum_{r \in \mathcal{R}} w_r K_r h_r(u)^2.} \quad (27)$$

Let

$$\Psi^\star := \min_{\|u\|=1} \Psi(u), \quad C_e^{\min} := \min_{\|u\|=1} C_e(u). \quad (28)$$

For a finite empirical sample, $C_e(u)$ is continuous in u , so both minima are attained.

Corollary 4.1 (Certificate-width factor). *Assume $C_e^{\min} > 0$ and define*

$$\rho_e(u_T) := \frac{C_e(u_T)}{C_e^{\min}} \geq 1. \quad (29)$$

Then

$$\boxed{\Psi(u_T) \leq \bar{\mu}_T R_M \rho_e(u_T)^2 \Psi^\star} \quad (30)$$

and, equivalently,

$$\boxed{C_e(u_T) G_\gamma(u_T) \leq \sqrt{\bar{\mu}_T R_M} \rho_e(u_T) \sqrt{\Psi^\star}.} \quad (31)$$

Proof. Theorem 3.3 gives

$$\Psi(u_T) = C_e(u_T)^2 J_\gamma(u_T) \leq C_e(u_T)^2 \bar{\mu}_T R_M J_\gamma^\star.$$

Moreover,

$$\Psi^\star = \min_u C_e(u)^2 J_\gamma(u) \geq (C_e^{\min})^2 \min_u J_\gamma(u) = (C_e^{\min})^2 J_\gamma^\star.$$

Combining the two displays proves (30); taking square roots gives (31). $\square$

Remark 4.2 (Scope of the comparator). Theorem 3.3 compares directions only through J_γ ; it does not by itself compare the full certificate-width objective Ψ . The distinction can be arbitrarily large. Consider one route in $\mathbb{R}^2$ with $K = 2$, $b_0 = e_1$, and $b_1 = e_2$. Let $X \sim \mathcal{N}(0, \text{diag}(\sigma_1^2, \sigma_2^2))$ and write $u = (\cos \phi, \sin \phi)$. For the population analogue $C_{e,\text{pop}}(u) := \text{CVaR}_\alpha(\|(I - uu^\top)X\|)$, define $\Psi_{\text{pop}}(u) := C_{e,\text{pop}}(u)^2 J_\gamma(u)$. If $c_\alpha := \text{CVaR}_\alpha(|Z|)$ for $Z \sim \mathcal{N}(0, 1)$, then

$$J_\gamma(u) = 2 \max\{\sin^2 \phi, \cos^2 \phi\}, \quad (32)$$

$$C_{e,\text{pop}}(u)^2 = c_\alpha^2 (\sigma_1^2 \sin^2 \phi + \sigma_2^2 \cos^2 \phi). \quad (33)$$

The direction $u_\gamma^\star = (e_1 + e_2)/\sqrt{2}$ attains $J_\gamma^\star = 1$, whereas

$$\frac{\Psi_{\text{pop}}(u_\gamma^\star)}{\Psi_{\text{pop}}^\star} \geq \frac{\Psi_{\text{pop}}(u_\gamma^\star)}{\Psi_{\text{pop}}(e_1)} = \frac{\sigma_1^2 + \sigma_2^2}{4\sigma_2^2},$$

which diverges as $\sigma_1/\sigma_2 \rightarrow \infty$. For $(\sigma_1, \sigma_2) = (10, 0.1)$ the lower bound is approximately 2500. Thus $J_\gamma^\star$ is the best single global direction for the slope-radius objective, not necessarily for the complete certificate half-width. Corollary 4.1 identifies the missing residual-direction factor ρ_e .

Lemma 4.3 (Tail-sensitive direction-uniform lower bound on residual CVaR). *Let $v_s = X_s - \mu$ and define*

$$\widehat{\Sigma}_X := \frac{1}{N} \sum_{s=1}^N v_s v_s^\top, \quad V_{\max} := \max_{1 \leq s \leq N} \|v_s\|, \quad B_X := \text{tr}(\widehat{\Sigma}_X) - \lambda_{\max}(\widehat{\Sigma}_X). \quad (34)$$

Assume $V_{\max} > 0$ and put $h = (1 - \alpha)N$. If $0 < \alpha < 1$, then for every unit direction u ,

$$\boxed{C_e(u) \geq L_\alpha := \frac{-V_{\max} h + \sqrt{V_{\max}^2 h^2 + 4(N - h) N B_X}}{2(N - h)}}. \quad (35)$$

For $\alpha = 0$, the limiting statement is

$$C_e(u) \geq L_0 := \frac{B_X}{V_{\max}}. \quad (36)$$

Consequently, L_α is a direction-uniform lower bound on $C_e^{\min}$.

Proof. Fix u and write $a_s = \|(I - uu^\top)v_s\| \geq 0$. Sort the values as $a_{[1]} \geq \dots \geq a_{[N]}$ and let $\omega_1, \dots, \omega_N \in [0, 1]$ be the fractional upper-tail weights, so that

$$\sum_{\ell=1}^N \omega_\ell = h, \quad C_e(u) = \frac{1}{h} \sum_{\ell=1}^N \omega_\ell a_{[\ell]}. \quad (37)$$

The nonzero weights occupy the largest order statistics. Hence every index with $1 - \omega_\ell > 0$ satisfies $a_{[\ell]} \leq C_e(u)$, including the fractionally weighted boundary value. Also $a_{[\ell]} \leq V_{\max}$ by contraction of the orthogonal projection. Therefore

$$\sum_{\ell=1}^N a_{[\ell]}^2 = \sum_{\ell=1}^N \omega_\ell a_{[\ell]}^2 + \sum_{\ell=1}^N (1 - \omega_\ell) a_{[\ell]}^2 \quad (38)$$

$$\leq V_{\max} \sum_{\ell=1}^N \omega_\ell a_{[\ell]} + (N - h) C_e(u)^2 \quad (39)$$

$$= V_{\max} h C_e(u) + (N - h) C_e(u)^2. \quad (40)$$

On the other hand,

$$\frac{1}{N} \sum_{s=1}^N a_s^2 = \text{tr}(\widehat{\Sigma}_X) - u^\top \widehat{\Sigma}_X u \geq B_X. \quad (41)$$

Combining the last two displays gives

$$(N - h) C_e(u)^2 + V_{\max} h C_e(u) - N B_X \geq 0. \quad (42)$$

When $0 < \alpha < 1$, the positive root of this quadratic is (35). When $\alpha = 0$, $h = N$ and the quadratic term vanishes, yielding (36). Every quantity on the right-hand side is independent of u . $\square$

Remark 4.4 (Relation to the mean-based bound). The earlier estimate $B_X/V_{\max}$ is exactly the $\alpha = 0$ specialization of Lemma 4.3. For $\alpha > 0$, inequality (40) retains the concentration of empirical CVaR on the largest $h = (1 - \alpha)N$ residual norms and can be substantially tighter.

5 Spectral transfer from training to search

Let $\lambda_1 \geq \dots \geq \lambda_d \geq 0$ be the eigenvalues of the search moment M . Let θ be the angle from u_T to the leading eigenspace $\mathcal{E}_1$:

$$\cos \theta = \|\Pi_{\mathcal{E}_1} u_T\|.$$

If $\theta = 0$, the transfer identity below is immediate and the term multiplied by $\sin^2 \theta$ is interpreted as zero. If $\theta > 0$, define

$$q = \frac{(I - \Pi_{\mathcal{E}_1})u_T}{\sin \theta}, \quad q \perp \mathcal{E}_1, \quad \|q\| = 1.$$

When $\theta = \pi/2$, no normalized leading-eigenspace component is required. If $\mathcal{E}_1 = \mathbb{R}^d$, then $\theta = 0$ for every u_T and the identity is trivial.

Proposition 5.1 (Exact spectral transfer identity). *For every fixed u_T ,*

$$\boxed{J_S(u_T) - J_S(u_S) = \sin^2 \theta (\lambda_1 - q^\top M q).} \quad (43)$$

Consequently,

$$J_S(u_T) \leq J_S(u_S) + (\lambda_1 - \lambda_d) \sin^2 \theta. \quad (44)$$

If $J_S(u_S) > 0$, then

$$R_M \leq 1 + \frac{(\lambda_1 - \lambda_d) \sin^2 \theta}{\text{tr}(M) - \lambda_1}. \quad (45)$$

Proof. Decompose u_T into its leading-eigenspace projection and orthogonal component. Because M acts as $\lambda_1 I$ on $\mathcal{E}_1$ and leaves its orthogonal complement invariant,

$$u_T^\top M u_T = \lambda_1 \cos^2 \theta + \sin^2 \theta q^\top M q.$$

Subtracting this quantity from $\text{tr}(M)$ and comparing with $J_S(u_S) = \text{tr}(M) - \lambda_1$ proves (43). Since $q^\top M q \geq \lambda_d$, inequality (44) follows. Division by $J_S(u_S)$ gives (45). $\square$

For $\theta > 0$ and $\lambda_1 > \lambda_d$, define the bottom-alignment statistic

$$\beta_q := \frac{q^\top M q - \lambda_d}{\lambda_1 - \lambda_d} \in [0, 1]. \quad (46)$$

When the additive bound is nonzero, it equals the fraction of that bound that is slack:

$$\frac{\Delta_{\text{bound}} - \Delta_{\text{exact}}}{\Delta_{\text{bound}}} = \beta_q, \quad (47)$$

where $\Delta_{\text{exact}} = J_S(u_T) - J_S(u_S)$ and $\Delta_{\text{bound}} = (\lambda_1 - \lambda_d) \sin^2 \theta$. Thus $\beta_q \approx 0$ means the component of u_T outside the leading eigenspace lies close to the bottom eigenspace.

Remark 5.2 (An a-priori ceiling on β_q). Let λ_{out} denote the largest eigenvalue of M on $\mathcal{E}_1^\perp$; when λ_1 is simple, $\lambda_{\text{out}} = \lambda_2$. Since $q \in \mathcal{E}_1^\perp$ is a unit vector, $q^\top M q \leq \lambda_{\text{out}}$, and hence

$$\beta_q \leq \frac{\lambda_{\text{out}} - \lambda_d}{\lambda_1 - \lambda_d}. \quad (48)$$

Because $M \succeq 0$, the right-hand side is at most $\lambda_{\text{out}}/\lambda_1$. Thus part of the near-exactness of the spectral bound is forced by a large leading eigengap. In the current diagnostic, $\lambda_2/\lambda_1 \leq 0.124$ while the measured β_q never exceeded 0.065; the remaining reduction reflects additional concentration of the orthogonal component toward the bottom of the spectrum.

6 Observed benchmark certificates

Two distinct experimental protocols must be kept separate.

Remark 6.1 (Protocol for the residual factor). The factor sample X is generated only from N , d , the support bound, and the random seed. These quantities were fixed, so the same $N \times d$ factor matrix was used for all 40 routing instances in the diagnostic. The sample consists of independent clipped standard-normal coordinates followed by centering and coordinatewise rescaling. Its empirical covariance was close to isotropic:

$$\frac{\lambda_{\max}(\widehat{\Sigma}_X)}{\lambda_{\min}(\widehat{\Sigma}_X)} = 1.0878, \quad \frac{\text{tr}(\widehat{\Sigma}_X)}{\lambda_{\max}(\widehat{\Sigma}_X)} = 9.613. \quad (49)$$

Consequently $C_e(\cdot)$ is one shared function of direction, not 40 independent instance-level functions. Its best-found directional spread was the single value

$$\frac{C_e^{\text{max,found}}}{C_e^{\text{min,found}}} = 1.0099. \quad (50)$$

Table 1: Current empirical evidence. The solver benchmark used $N = 50,000$; the geometry diagnostic used $N = 10,000$.

Quantity	Median	95th percentile	Worst observed
Solver speed-up ($N = 50,000$)	$2.92\times$	–	$1.62\times$ (minimum)
Certificate coverage ($N = 50,000$)	0.696	–	–
R_M ($N = 10,000$)	1.0205	1.0671	1.0790 (maximum)
$\bar{\mu}_T$ ($N = 10,000$)	3.0494	3.5875	4.4247 (maximum)
$\sqrt{\bar{\mu}_T} R_M$ ($N = 10,000$)	1.7587	1.9284	2.1061 (maximum)
$\lambda_2(M)/\lambda_1(M)$ ($N = 10,000$)	0.0935	0.1141	0.1236 (maximum)
$\angle(u_T, \mathcal{E}_1)$ ($N = 10,000$)	4.06°	6.21°	6.45° (maximum)
β_q ($N = 10,000$)	0.0264	0.0583	0.0648 (maximum)
$\hat{\rho}_e^{\text{found}}$ ($N = 10,000$)	1.00265	1.00286	1.00291 (maximum)
Best-found full-width factor ($N = 10,000$)	1.7635	1.9339	2.1117 (maximum)

The row for $\hat{\rho}_e^{\text{found}}$ in Table 1 does contain 40 observations because the learned direction u_T varies across instances. Thus the measured flatness of C_e reflects the nearly isotropic shared factor construction, not a property of the routing instances. Strongly anisotropic factor covariances can produce much larger variation, as the population example in the preceding scope remark shows.

The diagnostic supports the following interpretation.

1. All 40 realized search moments were numerically simple and strongly separated, with relative leading eigengap at least 0.876.
2. A direction learned from 200 random training routes increased the realized search-stream squared residual objective by at most 7.9%.
3. The observed residual coherence converted this surrogate guarantee into an instance-level weighted RMS exact-radius certificate of at most 2.11, with median 1.76.
4. The leading eigengap alone implies the a-priori ceiling $\beta_q \leq \lambda_2/\lambda_1 \leq 0.124$. The measured value was smaller still, $\beta_q \leq 0.065$, indicating additional bottom-spectrum alignment beyond that forced by the eigengap.
5. On the single shared, nearly isotropic factor sample, $C_e(u)$ varied by less than 1% across the searched directions. Across the 40 learned directions, $C_e(u_T)$ exceeded the smallest value found for that shared sample by at most 0.30%. Multiplying this residual factor by the radius factor changed the observed maximum from 2.1061 to a best-found full-width factor of 2.1117.

The last factor is descriptive rather than a rigorous global comparison: the numerical direction search provides an upper bound on $C_e^{\min}$ and therefore an optimistic estimate of ρ_e . For the shared factor sample,

$$\text{tr}(\widehat{\Sigma}_X) = 9.637798, \quad \lambda_{\max}(\widehat{\Sigma}_X) = 1.002552, \quad V_{\max} = 5.494648.$$

Here $\alpha = 0.9$, so $h = (1 - \alpha)N = 1000$. Lemma 4.3 therefore gives the rigorous direction-uniform bound

$$C_e^{\min} \geq \frac{-5.494648(1000) + \sqrt{5.494648^2(1000)^2 + 4(9000)(10000)(9.637798 - 1.002552)}}{2(9000)} \quad (51)$$

$$= 2.807281. \quad (52)$$

Since $L_\alpha \leq C_e^{\min}$, Corollary 4.1 gives the fully computable rigorous factor

$$F_{\text{rig}}(u_T) := \sqrt{\bar{\mu}_T R_M} \frac{C_e(u_T)}{L_\alpha}, \quad C_e(u_T) G_\gamma(u_T) \leq F_{\text{rig}}(u_T) \sqrt{\Psi^\star}. \quad (53)$$

Across the 40 learned directions, $F_{\text{rig}}(u_T)$ has median approximately 2.517, 95th percentile approximately 2.761, and maximum 3.014. For comparison, the $\alpha = 0$ mean-based specialization would give only 1.571574 and the previous maximum factor 5.384. On the shared sample, the best-found minimum 4.0072 is about 1.43 times the strengthened rigorous lower bound. The remaining gap reflects the uniform tail relaxation, not observed directional variability in C_e .

These are observed population certificates, not universal constants. In particular, the distribution-free bound is only $\mu_r(u) \leq K_r$, and this bound is tight.

7 What is proved, and what remains open

The present theorem stack gives:

- an exact routewise CVaR bracket for any fixed direction;
- exact decisions through full fallback;
- an exact second-moment characterization of the SLOPE direction;
- a direction-free sufficient margin connecting route half-width to certification;
- a jointly tight deterministic bound relative to the best single global direction for the slope-radius objective;
- a rigorous certificate-width corollary that exposes the additional residual factor ρ_e ;
- an exact spectral identity for training-to-search transfer.

It does *not* give:

- a universal constant such as 2.11;
- a comparison with independent per-route oracle directions;
- a routewise approximation guarantee from the aggregate RMS theorem;
- an a-priori probabilistic guarantee that a training moment will transfer to a future search stream;
- a universal small bound on the residual-direction factor ρ_e or a guarantee that the SLOPE direction minimizes the complete certificate half-width;
- a guarantee that minimizing the aggregate objective Ψ exactly maximizes finite-population coverage;
- a structural bound on coherence stronger than $\mu_r \leq K_r$ without additional assumptions.

The next empirical step is to test whether the same mechanism persists beyond the current Dethloff affine overlay, beginning with anisotropic factor covariances and then varying dimensions, uncertainty profiles, route-normalized weighting, and a second max-affine application family. This note is intended as a clean mathematical audit target before those generalizations are added.

A Degenerate cases

1. If $M = 0$, then every slope is zero and $J_S \equiv J_\gamma \equiv 0$. The ratio $R_M = 0/0$ is undefined and should not be reported.
2. If $M \neq 0$ but $J_S(u_S) = 0$, then M is rank one and $J_\gamma^\star = 0$. If u_T belongs to the leading eigenspace, then $J_\gamma(u_T) = 0$. Otherwise no finite multiplicative ratio relative to $J_\gamma^\star$ exists; only the additive spectral bound remains meaningful.
3. If $S_r(u_T) = 0$ for one route, then $\gamma_r(u_T) = 0$ and that route contributes zero to both sides of the coherence step.
4. Repeated route checks are represented by larger weights w_r ; the proofs are unchanged.
5. Taking $w_r = 1$ gives the current cut-weighted SLOPE objective. Taking $w_r = 1/K_r$ gives a route-normalized variant with $J_\gamma(u) = \sum_r \gamma_r(u)^2$.